Deploying Nhill Server on a Hostinger VPS (France)

Français : ../fr/deploiement-vps-hostinger.md

This guide installs Nhill Server and its PostgreSQL database on **a Hostinger VPS located in France**, running Ubuntu 24.04, then covers operations: backups, restore, monitoring, updates and rollback. Commands run on the VPS unless stated otherwise.

> **Verification status (28 October 2026).** The full procedure (`deploy/vps/provision.sh` then `deploy/vps/tests/acceptance.sh`) is run by the CI job `vps-test` on a fresh Ubuntu 24.04 machine, with a certificate from Caddy's local authority. It has **not** yet been run on the production Hostinger VPS, nor with a real Let's Encrypt certificate: run it on a staging VPS first.

1. Architecture

```
Applications (Business, POS, Console)
        │ HTTPS 443 (TLS 1.2+, Let's Encrypt certificate renewed by Caddy)
        ▼
Hostinger firewall (before the VPS)  →  ufw (on the VPS)
        ▼
Caddy :80/:443 ──► nhill-server 127.0.0.1:8080 ──► PostgreSQL 16 127.0.0.1:5432
                                                          │
        daily encrypted backup (age) ◄────────────────────┘
        /var/backups/nhill/archives  ──►  off-site copy (outside Hostinger)
```

Choice	Reason
One VPS, local database	modest initial volume; the database is never exposed to the network (listens on 127.0.0.1 and ::1 only)
Caddy in front of the API	automatic TLS (ACME), security headers, request size limit; the API listens locally only
Separate PostgreSQL roles	`nhill_owner` (migrations), `nhill_app` (API, cannot bypass row-level security), `nhill_backup` (read-only)
Backups encrypted with `age`	only the **public** key is on the VPS: stealing the VPS or the backups does not reveal the data
Database copy before every update	rollback in one command, without the backup private key
Data in France (European Union)	location chosen when the VPS is created; it cannot change afterwards

Hosting provider. For a customer established in the DRC, Hostinger's contracting entity is Hostinger International Ltd, 61 Lordou Vironos str., 6023 Larnaca, Cyprus (Hostinger's terms of service, § 2, and their list of countries). Hostinger offers a data processing agreement (DPA) listing its sub-processors; accepting and keeping it belongs to the compliance file (record of processing, legal template 18).

2. Ports, flows and firewall

Port	Listener	Open to the Internet	Role
22/tcp (or your SSH port)	sshd	yes, preferably restricted to your addresses	administration (SSH keys)

Port	Listener	Open to the Internet	Role
80/tcp	Caddy	yes	Let's Encrypt ACME challenge, redirect to HTTPS
443/tcp, 443/udp	Caddy	yes	API over HTTPS (HTTP/2, HTTP/3)
8080/tcp	nhill-server	**no** (127.0.0.1)	API behind Caddy
5432/tcp	PostgreSQL	**no** (127.0.0.1, ::1)	database

Two complementary firewalls:

1. **Hostinger managed firewall** (VPS panel → *Security* → *Firewall*). It filters before traffic reaches the VPS; an empty rule group **blocks everything**. Create a group with "accept" rules for 22/tcp (source: your addresses if possible), 80/tcp, 443/tcp and 443/udp, then activate it. Changes apply to new connections within two minutes.
2. **ufw on the VPS**, set by `provision.sh`: all incoming traffic refused except SSH (rate limited), 80/tcp and 443/tcp+udp. `fail2ban` bans an address for one hour after 5 SSH failures within 10 minutes.

3. Before you start

1. **Create the VPS** at Hostinger: location **France**, **Ubuntu 24.04** plain OS (no panel), 2 vCPU and 4 GB RAM recommended (2 GB minimum), an SSH key added at creation.
2. **DNS**: an `A` record (and `AAAA` if the VPS has IPv6) for the API name, pointing to the VPS. This guide uses `api.nhill.online`, a subdomain of the official domain; another name works if it is the same everywhere. The certificate is issued only once that name resolves to the VPS.
3. **Backup key, on your computer** (never on the VPS): `age-keygen -o nhill-backup-key.txt`; note the public key it prints (`age1…`); keep the file in your password safe and an offline copy. Without it, no backup can be read.
4. **Package**: `nhill-server_3.0.0-1_amd64.deb` and its `SHA256SUMS` (`dist/linux/ubuntu-24.04/` or the release), plus the repository's `deploy/vps/` folder (also installed by the package in `/usr/share/nhill/vps/`).
5. **Off-site copy** (recommended): storage outside Hostinger, in the European Union, and a copy command (for example `rclone` configured for the `nhill-backup` user).

4. Installation

The commands read four values specific to your installation, to set first in the terminal of your computer: the VPS IP address (Hostinger panel), the backup public key (step 3), the e-mail address of the first Nhill Console administrator and their name.

```
export VPS_IP=…            # IPv4 address of the VPS, shown by Hostinger
export BACKUP_KEY=age1…    # public key printed by age-keygen (step 3)
export ADMIN_EMAIL=…       # address of the first Nhill Console administrator
export ADMIN_NAME=…        # their name, shown in the console
scp -r nhill-server_3.0.0-1_amd64.deb SHA256SUMS deploy/vps root@"$VPS_IP":/root/
ssh -t root@"$VPS_IP" "cd /root && ./vps/provision.sh \
  --deb nhill-server_3.0.0-1_amd64.deb --sha256sums SHA256SUMS \
  --api-domain api.nhill.online --acme-email info@nhill.online \
  --admin-email '$ADMIN_EMAIL' --admin-name '$ADMIN_NAME' \
  --backup-recipient '$BACKUP_KEY' \
  --support-email support@nhill.online \
  --offsite-cmd 'rclone copy {} remote:nhill-backups/' \
  --harden-ssh"
```

Without an off-site copy, remove the `--offsite-cmd` line.

The script stops at the first error; re-run, it keeps what is installed. It:

1. checks the system (Ubuntu 24.04, amd64, memory, disk) and **the package SHA-256**;
2. installs from the Ubuntu archive: PostgreSQL 16, Caddy, age, ufw, fail2ban and automatic security updates (no automatic reboot);
3. configures ufw and fail2ban; with `--harden-ssh`, accepts SSH keys only (ignored while no key is installed, so that you are not locked out);
4. restricts PostgreSQL to the loopback and creates the `nhill` database;
5. installs the package, then the database with `nhill-server setup` through a temporary superuser **deleted immediately** (its password only goes through standard input); checks the isolation (10 checks);
6. sets up the encrypted backups and takes a first one;
7. writes `/etc/caddy/Caddyfile` for `api.nhill.online` (Let's Encrypt);
8. installs the health checks, log retention and the `nhill-update`, `nhill-rollback` tools;
9. prints the **one-time token** of the first Nhill Console administrator (30 minutes).

Then:

- **Back up `NH_SECRET_KEY`** (`sudo grep '^NH_SECRET_KEY=' /etc/nhill/server.env`) in your password safe, apart from the backups: without it, establishment codes and administrators' second factors can no longer be read.
- **Nhill Console**: address `https://api.nhill.online`, "First access", token, password (14 characters minimum) and TOTP. Token expired or lost, while nobody has completed the first access: `sudo -u nhill nhill-server --config /etc/nhill/server.env admin-bootstrap <same-address> "<Name>"`.
- **E-mails**: set the SMTP relay (`NH_EMAIL_TRANSPORT=smtp`, `NH_SMTP_*`, configuration guide) with a sender address of the domain authorised by SPF and DKIM, then `sudo systemctl restart nhill-server`. While `NH_EMAIL_TRANSPORT=log`, no e-mail (recovery codes, reminders) is sent.

5. Secrets

Secret	Where	Permissions	Backup
`NH_SECRET_KEY` (master key)	`/etc/nhill/server.env`	600, `nhill`	password safe, kept apart
`nhill_owner`, `nhill_app` passwords	`/etc/nhill/server.env`	600, `nhill`	not needed (regenerated by `nhill-server setup`)

Secret	Where	Permissions	Backup
`nhill_backup` password	`/var/backups/nhill/.pgpass`	600, `nhill-backup`	not needed
age **public** key	`/etc/nhill/backup.env`	600, `nhill-backup`	—
age **private** key	**outside the VPS**	—	safe + offline copy
SMTP password	`/etc/nhill/smtp-password` (`NH_SMTP_PASSWORD_FILE`)	600, `nhill`	safe
Payment provider secrets	database, encrypted with `NH_SECRET_KEY`	—	backups

No secret is in the repository, the package or the logs.

6. Certificates

Caddy obtains the certificate of `api.nhill.online` from Let's Encrypt (HTTP-01 challenge on port 80) and renews it by itself, about 30 days before expiry. The health check alerts when fewer than 14 days remain (`NH_HEALTH_CERT_MIN_DAYS` in `/etc/nhill/health.env`; 0 with `--tls internal`, whose short-lived certificates Caddy renews itself). On failure: `journalctl -u caddy`, DNS resolution of the name, ports 80 and 443 open in **both** firewalls.

7. Backups

Level	Frequency	Retention	Protects against
Encrypted Nhill backup (`nhill-backup.timer`)	daily around 02:30 UTC (+0 to 15 min)	14 days on the VPS	mistakes, deletion, database corruption
Off-site copy (`NH_BACKUP_OFFSITE_CMD`)	every backup	per your storage (30 days or more advised)	loss of the VPS or of the Hostinger account
Hostinger backups of the whole VPS	weekly by default; daily as a paid option; 2 weekly and 2 daily kept	—	whole-system recovery; **a restore overwrites the whole VPS**
Hostinger snapshot	on demand (before an update)	one, deleted after 1 day	whole-system rollback right after an operation
Database copy before update (`nhill_pre_…`)	every `nhill-update`	last 2	rollback of an update

Hostinger facts from its support article "How to back up or restore a VPS", read on 28 October 2026. Hostinger backups **complement** the Nhill backups and do not replace them: they are neither encrypted by you nor kept outside Hostinger.

Check: `systemctl list-timers nhill-backup.timer`, `ls -lh /var/backups/nhill/archives`, `journalctl -u nhill-backup`.

8. Restore

Monthly test, outside the VPS (administrator's computer or staging VPS, with the private key):

```
ssh root@"$VPS_IP" ls -1t /var/backups/nhill/archives | head -3   # latest backups
export BACKUP=…   # name of a backup from the list, for example the latest
scp root@"$VPS_IP":"/var/backups/nhill/archives/$BACKUP*" .
deploy/vps/nhill-restore-test.sh "$BACKUP" nhill-backup-key.txt
```

The script starts a throw-away PostgreSQL (local socket, nothing on the network), restores, counts migrations, establishments, users, documents and audit entries, checks forced row-level security and the isolation of the application role, then deletes everything.

Real restore on the VPS (after data loss):

1. Copy the backup (with its `.sha256`) and, **temporarily**, the private key into `/var/lib/postgresql/restore/` (owner `postgres`, mode 700).
2. `cd /var/lib/postgresql/restore && sudo -u postgres env PGHOST=/var/run/postgresql nhill-restore.sh "$BACKUP" key.txt nhill_restored` (restores into a **new** database, then shows migrations, establishments, row-level security).
3. `sudo systemctl stop nhill-server`, then rename: `ALTER DATABASE nhill RENAME TO nhill_before_restore; ALTER DATABASE nhill_restored RENAME TO nhill;` and give the database privileges again: `GRANT CONNECT, CREATE ON DATABASE nhill TO nhill_owner; GRANT CONNECT ON DATABASE nhill TO nhill_app, nhill_backup;`
4. `sudo systemctl start nhill-server`, check `nhill-healthcheck`, then **delete the private key from the VPS** (`sudo shred -u /var/lib/postgresql/restore/key.txt`).

Data entered between the backup and the incident is lost: at most one day with the default frequency.

9. Monitoring and logs

`nhill-healthcheck.timer` checks every 5 minutes: services `nhill-server` and `caddy`; PostgreSQL answering (`pg_isready`); `/health` locally and over HTTPS through Caddy; certificate expiry; ports 5432 and 8080 on loopback only; ufw active; disk (< 85 %); last backup younger than 26 hours. Result: `/var/lib/nhill/health.json`. A failure starts `nhill-alert`: a message in the system journal (priority `err`) and, when `NH_ALERT_CMD` is set in `/etc/nhill/alert.env`, your alert command (e-mail, chat, notification).

A VPS cannot report its own total outage: add an **external probe** on `https://api.nhill.online/health` (uptime monitoring service of your choice).

Log	Where	Retention
API (JSON per line)	`journalctl -u nhill-server`	90 days, 1 GB at most (journald)
HTTPS access (IP addresses)	`/var/log/caddy/nhill-access.log`	10 × 50 MB, 90 days at most
PostgreSQL	`/var/log/postgresql/`	Ubuntu rotation (logrotate)
Backups, health	`journalctl -u nhill-backup -u nhill-healthcheck`	90 days
Application errors	Nhill Console → Technical errors	database

10. Updates and rollback

```
sudo nhill-update nhill-server_3.0.1-1_amd64.deb --sha256sums SHA256SUMS
```

1. checks the package (name, checksum, newer version);
2. takes an encrypted backup (and its off-site copy);
3. stops the API and copies the database (`nhill_pre_<version>_<date>`);
4. installs the package; at start, the server applies the new migrations;
5. waits for `/health`; on failure, **automatic rollback**.

Downtime: from step 3 to the end of step 5, usually under a minute for a small database. Before an important update, also take a **Hostinger snapshot**.

Manual rollback (for example after a defect found later): `sudo nhill-rollback` reinstalls the previous package and the database copy; the database used since the update is kept as `nhill_failed_<date>` (entries made since then are there).

Migrations: versioned (`db/migrations/NNNN_*.sql`), applied in order, one transaction each, under a lock; a migration changed after being applied is refused (checksum). They only go forward: an older server **refuses to start** on a database migrated by a newer version (message in `journalctl -u nhill-server`); going back therefore uses the database copy, never a "down" migration.

System updates: security fixes are automatic; schedule a reboot when `/var/run/reboot-required` exists (`sudo reboot`, the API restarts by itself).

11. Reinstalling on a new VPS (recovery plan)

1. Create a VPS (France, Ubuntu 24.04), DNS, Hostinger firewall: §§ 2-3.
2. Run `provision.sh` (§ 4) with the same domain.
3. Put the original `NH_SECRET_KEY` back in `/etc/nhill/server.env` (safe), then restore the latest off-site backup (§ 8, real restore).
4. `sudo systemctl restart nhill-server`, `nhill-healthcheck`, test sign-in from Nhill Console and Nhill Business.

12. Personal data and compliance

- The data of the businesses and of their customers is hosted in France, in the European Union; Hostinger's contract and DPA govern this processor.
- Businesses located outside the European Union (for example in the DRC) transfer their data to France: the obligations of their country (declaration, transfer authorisation) are listed in `docs/en/legal-parameters.md` and `legal/en/00-legal-research.md`. No compliance is claimed without review by a lawyer.
- Limited log retention (90 days); encrypted backups; SSH key access to the VPS.

13. Left to the operator

Action	Where
Accept Hostinger's DPA and file it	Hostinger account
Configure the Hostinger firewall	VPS panel
Choose and configure the off-site copy	`--offsite-cmd`, `rclone`

Action	Where
Configure the SMTP relay and SPF/DKIM/DMARC of the sender domain	`server.env`, DNS zone
Add an external uptime probe	service of your choice
Set `NH_ALERT_CMD`	`/etc/nhill/alert.env`
Run the first restore test outside the VPS, then monthly	§ 8